Summary of Advanced Merger Model Features

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Overview of Advanced Merger Model Features

This guide covers **features common in more advanced merger models**, such as quarterly projections, Exchange Ratios with collars, and Stock vs. Asset vs. 338(h)(10) deals.

We assume you already know the fundamentals, so we do not spend time re-explaining the basic assumptions or model structure (please see the introductory guide for that coverage).

This guide is less of a step-by-step walkthrough and more of a "quick reference" for formulas and schedules in merger models. You can jump to specific topics using the Table of Contents above.

Quarterly Merger Models

In a standard merger model, the deal closes at the end of the Buyer's fiscal year, and the Buyer and Seller have the *same* fiscal years (e.g., ones that end on December 31st).

In real life, both assumptions are often false:

- The Buyer's fiscal year might end on June 30th, but the Seller's might end on Dec. 31st.
- The deal might close on March 31st, September 30th, or another date that does not coincide with the end of a quarter or year.

The first problem is easier to deal with: **Always use the Buyer's fiscal year in merger models**.

If the Buyer's fiscal year ends on June 30th but the Seller's ends on December 31st, you can take the results from the last six months of the Seller's fiscal year and add them to the first six months of the next year.

You then add those results to the ones from the Buyer's fiscal year since they now cover the same period (January 1st – December 31st).

If the merger model is built on a **quarterly basis**, you can use functions like XLOOKUP and INDEX/MATCH to retrieve the financials for the corresponding dates and add together quarters.

The second problem with irregular closing dates is more cumbersome to deal with.

If the deal closes between fiscal years or quarters, you must create **stub periods** in the model.

For example, if the deal closes on August 14th, you'd have to create two stub periods in a quarterly model: One that shows the Buyer and Seller's results from July 1st to August 14th and another that shows their results from August 15th to September 30th.

To do this, you would multiply the Income Statement and Cash Flow Statement line items by the percentages of the quarter that the stub periods represent.

July 1st to August 14th is 45 days inclusive, so this is 45 / 92 = 48.9%, and August 15th to September 30th is 47 days inclusive, which is 47 / 92 = 51.1%.

You would also have to create a Balance Sheet forecast for August 14th based on the July 1st to August 14th stub period, with links for each "pairing" (e.g., AR on August 14th would be based on Revenue over this period).

The first "full" combined quarter would run from October 1st to December 31st, and you would focus on the first full combined year for analytical purposes.

Quarterly Model Setup and Projections

The main differences in quarterly 3-statement models are as follows:

- **Year-Over-Year (YoY) Growth Rates:** Instead of projecting the growth from quarter to quarter, such as Q3 to Q4, you should use *Year-Over-Year* growth rates, such as Q4 of 20X5 vs. Q4 of 20X4. This is due to the seasonality in many businesses:

	A B	C	D	E	P	Q	R	S	T
100									
101									
102		**Target - Income Statement:**	***Units:***		**3Q 20**	**4Q 20**	**1Q 21**	**2Q 21**	**3Q 21**
103		**Revenue:**							
104		(+) Structural Components:	*$ M*		$ 201.6	$ 173.1	$ 178.0	$ 163.7	$ 221.6 $
105		(+) Lumber & Lumber Sheet Goods:	*$ M*		317.5	275.6	288.9	291.3	=+P105*(1+T42)
106		(+) Windows, Doors & Millwork:	*$ M*		297.2	286.4	307.5	296.6	318.0
107		(+) Other Building Products & Services:	*$ M*		255.6	233.7	212.0	238.1	271.2
108		**Total Revenue:**	*$ M*		**1,072.0**	**968.8**	**986.4**	**989.7**	**1,159.6**

- **Balance Sheet Projections:** Anything based on an Income Statement figure, such as Accounts Receivable, should use the *Last Twelve Months (LTM)* figure on the IS rather than a single quarter's numbers. For example, AR for Q3 of 20X5 should be based on the LTM Revenue between Q4 of 20X4 and Q3 of 20X5.

	A B	C	D	E	P	Q	R	S	T
100									
101									
102		**Target - Income Statement:**	***Units:***		**3Q 20**	**4Q 20**	**1Q 21**	**2Q 21**	**3Q 21**
103		**Revenue:**							
104		(+) Structural Components:	*$ M*		$ 201.6	$ 173.1	$ 178.0	$ 163.7	$ 221.6
105		(+) Lumber & Lumber Sheet Goods:	*$ M*		317.5	275.6	288.9	291.3	348.8
106		(+) Windows, Doors & Millwork:	*$ M*		297.2	286.4	307.5	296.6	318.0
107		(+) Other Building Products & Services:	*$ M*		255.6	233.7	212.0	238.1	271.2
108		**Total Revenue:**	*$ M*		**1,072.0**	**968.8**	**986.4**	**989.7**	**1,159.6**
109		(-) Cost of Sales:	*$ M*		(775.0)	(698.5)	(721.1)	(723.5)	(847.7)
110		**Gross Profit:**	*$ M*		**296.9**	**270.3**	**265.3**	**266.2**	**311.9**
131									
132									
133		**Target - Balance Sheet:**	***Units:***		**3Q 20**	**4Q 20**	**1Q 21**	**2Q 21**	**3Q 21**
134		**ASSETS:**							
135		**Current Assets:**							
136		Cash & Cash Equivalents:	*$ M*		$ 197.7	$ 188.8	$ 208.1	$ 268.7	$ 234.1
137		Accounts Receivable:	*$ M*		355.4	350.8	363.9	=+S65*SUM(P108:S108)	
138		Inventory & Other Current Assets:	*$ M*		459.1	444.1	473.9	458.5	484.9
139		**Total Current Assets:**	*$ M*		**1,012.3**	**983.7**	**1,045.8**	**1,110.2**	**1,113.0**

- **Debt, Cash, and Interest:** Anything based on an *annual* rate or repayment percentage must be divided by 4 in a quarterly model (there is a more precise way to convert annual rates to quarterly rates, but this is fine in most models). If the annual interest rate on Debt is 8%, the *quarterly interest* should be 2% * Quarterly Debt Balance.

Cash & Debt Schedule:		3Q 20	4Q 20	1Q 21	2Q 21	3Q 21	4Q 21
Debt Maturities & Repayments:	$ M	52.8	-	62.5	62.5	62.5	62.5
Issuances % Maturities & Repayments:	%	70.0%	90.0%	90.0%	90.0%	90.0%	90.0%
Debt Issuances:	$ M	37.0	-	56.3	56.3	56.3	56.3
Cash Interest Payments:	$ M	4.8	4.7	5.0	=+S92*AVERAGE(R91,S91)/Num_Periods		
Face Value of Total Debt Balance:	$ M	346.7	346.7	340.4	334.2	327.9	321.7
Weighted Average Interest Rate (Annualiz	%	5.4%	5.4%	5.8%	5.8%	5.8%	5.8%

- **Credit Stats and Ratios:** If you are mixing Balance Sheet and Income Statement figures, such as Debt / EBITDA, you must use the *LTM* version of the Income Statement metric. So, it's always Debt At the End of the Quarter / LTM EBITDA.

	A	B	C	D	E	F	G	H	I	J
227			**New Debt Balance - Face Value:**	*$ M*		-	-	-	-	-
228			**Total Debt Balance - Face Value:**	*$ M*		**1,959.2**	**1,985.9**	**1,858.7**	**1,577.1**	**1,594.1**
229										
230										**Historical:**
231			**Combined Company - Key Metrics and Ratios:**	***Units:***		**1Q 18**	**2Q 18**	**3Q 18**	**4Q 18**	**1Q 19**
232										
233			**Operating Income:**	*$ M*		**$ 52.1**	**$ 104.6**	**$ 121.8**	**$ 90.5**	**$ 71.9**
234			(+) Depreciation:	*$ M*		17.3	18.7	19.1	19.3	19.8
235			(+) Amortization of Intangibles:	*$ M*		5.5	6.0	6.0	6.0	3.8
236			(+) Amortization of New Intangibles:	*$ M*		-	-	-	-	-
237			(+) Depreciation of PP&E Write-Up:	*$ M*		-	-	-	-	-
238			(+) Impairment Charges:	*$ M*		-	-	-	-	-
239			**EBITDA:**	*$ M*		**75.0**	**129.3**	**146.9**	**115.7**	**95.5**
240										
241			LTM EBITDA:	*$ M*					466.9	487.4
242										
253			*Total Debt / LTM EBITDA:*	*x*					=+I228/I241	*3.3 x*
254			*Net Debt / LTM EBITDA:*	*x*					*3.4 x*	*3.2 x*
255			*LTM EBITDA / Net Interest Expense:*	*x*					*4.3 x*	*4.6 x*

- **Annual Roll-Ups and Summaries:** Finally, it helps to create annual roll-ups that summarize the Income Statement and Cash Flow Statement with SUMIF and the Balance Sheet with INDEX/MATCH or XLOOKUP. Otherwise, it's difficult to interpret all the numbers.

	C	D	E	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN
100																			
101										**Historical:**		**Projected:**							
102	**rget - Income Statement:**	***Units:***		**4Q 22**	**1Q 23**	**2Q 23**	**3Q 23**	**4Q 23**		**FY18**	**FY19**	**FY20**	**FY21**	**FY22**	**FY23**				
103	**Revenue:**																		
104	(+) Structural Components:	*$ M*		194.6	$ 205.7	$ 162.2	$ 253.2	$ 204.3		$ 622.1	$ 636.0	$ 699.6	$ 748.6	$ 786.1	=SUMIF($F$35:$AC$35,AJ$35,$F104:$AC104)				
105	(+) Lumber & Lumber Sheet Goods:	*$ M*		310.9	335.8	289.0	399.2	326.7		1,286.5	1,040.9	1,145.0	1,225.1	1,286.4	1,350.7				
106	(+) Windows, Doors & Millwork:	*$ M*		314.3	323.1	308.6	343.3	325.3		964.7	1,080.8	1,167.3	1,225.7	1,262.4	1,300.3				
107	(+) Other Building Products & Services:	*$ M*		254.6	222.5	247.1	292.5	263.6		809.2	868.8	929.7	966.8	995.9	1,025.7				
108	**Total Revenue:**	*$ M*		**1,074.4**	**1,087.0**	**1,006.8**	**1,288.3**	**1,120.0**		**3,682.4**	**3,626.6**	**3,941.6**	**4,166.4**	**4,330.8**	**4,502.1**				

The SUMIF function here is based on the numerical year numbers in row 35, which is offscreen; we add only the quarters whose year numbers match those of the full fiscal years.

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Variable Close Dates

Variable close dates affect the model in two main ways:

1. **Deal Financing and Assumptions** – Depending on the close date, the deal's Cash, Debt, and Stock mix might change as the Acquirer and Target's Cash and Debt levels change over time.

2. **Financial Statement Combination** – You must use INDEX/MATCH or XLOOKUP to bring in the companies' separate financials and match them to the dates. This could produce very different accretion/dilution figures when measured annually.

With point #1, when calculating the Cash and Debt available for use in a deal, you must bring in the numbers from the Acquirer and Target *on the close date* and *calculate their LTM EBITDAs on the close date*. Here are a few examples:

	C	D	E
39			
40	Post-Transaction Minimum Cash Balance:	*$ M*	$ 300.0
41	Post-Transaction Combined Cash Balance:	*$ M*	=INDEX(Acquirer!$F$127:$AC$127,1,
42	Maximum Cash Available for Deal:	*$ M*	MATCH(Close_Date,Acquirer!$F$124:$AC$124,0))
43			+INDEX(Target!$F$136:$AC$136,1,
44			MATCH(Close_Date,Target!$F$133:$AC$133,0))

The Post-Transaction Minimum Cash stays the same regardless of the close date, so the Maximum Cash Available for the Deal changes as the close date changes. We use INDEX/MATCH to find the Acquirer and Target's numbers based on the assumed close date.

The Maximum Debt setup is similar but requires two separate formulas to bring in the Debt and the LTM EBITDA on the close date (the Debt part is shown below):

	C	D	E
43			Target - LTM EBIT
44	Maximum Combined LTM Debt / EBITDA:	*x*	4.0 x
45	Combined Debt on Close Date:	*$ M*	=INDEX(Acquirer!$F$83:$AC$83,1,
46	Combined LTM EBITDA on Close Date:	*$ M*	MATCH(Close_Date,Acquirer!$F$77:$AC$77,0))
47			+INDEX(Target!$F$91:$AC$91,1,
48	Maximum New Debt Available for Deal:	*$ M*	MATCH(Close_Date,Target!$F$85:$AC$85,0))
49	New Debt - Annual Amortization:	*%*	1.0%
50	New Debt - Maturity:	*# Years*	5
51	New Debt - Maturity Date:	*Date*	2027-06-30

Assuming the Acquirer and Target both grow modestly, *later close dates tend to improve EPS accretion* because they result in higher Cash balances and lower Debt balances.

Therefore, more Cash and Debt can fund the deal, which boosts EPS accretion since they are both cheaper than Stock.

To combine the financial statements, you can use INDEX/MATCH or XLOOKUP functions that reference just the Acquirer's numbers or add the Acquirer and Target's numbers, depending on whether the deal has closed:

Fiscal Year:			**2020**	**2020**	**2021**	**2021**
Combined Year #:		**0.00**	**0.00**	**0.00**	**0.25**	**0.50**
Combined Income Statement:	***Units:***		**3Q 20**	**4Q 20**	**1Q 21**	**2Q 21**
Revenue:						
(+) Lumber & Lumber Sheet Goods:	*$ M*		$ 720.5	$ 627.0	$ 869.0	$ 938.3
(+) Manufactured Products:	*$ M*		477.5	426.2	375.7	380.4
(+) Windows, Doors & Millwork:	*$ M*		448.1	453.7	741.9	716.9
(+) Structural Components:	*$ M*		-	-	178.0	163.7
(+) Gypsum, Roofing & Insulation:	*$ M*		187.1	141.7	118.6	134.0
(+) Siding, Metal & Concrete Products:	*$ M*		213.1	179.7	179.0	213.6
(+) Other Building Products & Services:	*$ M*		215.1	199.2	434.6	479.0
(+) Revenue Synergies:	*$ M*		-	-	59.6	58.8
Total Revenue:	*$ M*		**2,261.5**	**2,027.5**	**2,956.5**	**3,084.7**
(-) Cost of Sales:	*$ M*		(1,616.9)	(1,476.0)	(2,111.9)	(2,205.8)
(-) Cost of Revenue Synergies:	*$ M*		-	-	(43.6)	(43.0)
Total Cost of Sales:	*$ M*		**(1,616.9)**	**(1,476.0)**	**(2,155.5)**	**(2,248.8)**
Gross Profit:	*$ M*		**644.5**	**551.5**	**801.0**	**835.9**

We add the Target's separate financials only when the "Combined Year #" is greater than 0.00, indicating the deal has closed.

Many items must be recalculated on a quarterly basis on the Combined Income Statement and Cash Flow Statement; examples are the new D&A on asset write-ups, the Foregone Interest on Cash, and the Interest Paid on New Debt:

Fiscal Year:			**2020**	**2020**	**2021**	**2021**	**20**
Combined Year #:		**0.00**	**0.00**	**0.00**	**0.25**	**0.50**	**0.**
Combined Income Statement:	***Units:***		**3Q 20**	**4Q 20**	**1Q 21**	**2Q 21**	**3Q**
Operating Income:	*$ M*		**147.3**	**72.0**	**117.3**	**194.0**	
(-) Interest Expense:	*$ M*		(28.3)	(28.7)	(33.7)	(32.5)	
(+) Interest Income:	*$ M*		1.9	2.1	3.0	2.9	
(-) Foregone Interest on Cash:	*$ M*		=IF(P$89>0,				
(-) Interest Paid on New Debt Issued:	*$ M*		-($E$56+$E$60)*Cash_Foregone_Interest_Rate/Num_Periods,0)				
(-) Amortization of New Debt Issuance Fees	*$ M*		-	-	(1.2)	(1.2)	
(+) Other Income, Net:	*$ M*		-	-	1.8	2.4	
(-) Goodwill Impairment:	*$ M*		-	-	-	-	
Pre-Tax Income:	*$ M*		**120.9**	**45.4**	**60.2**	**138.5**	

The **Balance Sheet** is trickier to combine because there are three cases:

1. **Close Date** – You link in a "special version" of the Balance Sheet, which reflects the adjustments for Cash, Debt, Stock, Goodwill, other asset write-ups, etc.

2. **Pre-Close Date** – In this case, it's just the Acquirer's Balance Sheet, brought in via an INDEX/MATCH or XLOOKUP function.

3. **Post-Close Date** – Depending on the item, these are either additions of the Acquirer and Target's Balance Sheets (e.g., for the Working Capital lines) or they are based on the *previous quarter* plus/minus the corresponding changes on the Cash Flow Statement in this quarter (e.g., Net PP&E and CapEx and D&A on the CFS).

The formulas get long and complicated and are better understood in Excel, but here's an example of these three cases for the Combined Company's Net PP&E:

	A B	C	D	E	P	Q	R	S	T	U
149										
150		**Combined Balance Sheet:**			**3Q 20**	**4Q 20**	**1Q 21**	**2Q 21**	**3Q 21**	**4Q 21**
151		**ASSETS:**								
152		**Current Assets:**								
153		Cash & Cash Equivalents:	*$ M*		$ 420.7	$ 241.4	$ 199.0	$ 238.8	$ 171.8	$ 155.5
154		Accounts Receivable:	*$ M*		791.3	1,120.9	1,150.6	1,228.4	1,249.7	1,175.7
155		Inventory & Other Current Assets:	*$ M*		661.5	1,113.1	1,195.5	1,168.1	1,199.5	1,183.5
156		**Total Current Assets:**	*$ M*		**1,873.5**	**2,475.4**	**2,545.1**	**2,635.3**	**2,621.0**	**2,514.7**
157										
158		**Long-Term Assets:**								
159		Net Plants, Property & Equipment:	*$ M*		801.5	1,313.0	=IF(R$150=Close_Date,Combined_BS!$I15,			
160		Operating Lease Assets:	*$ M*		309.8	477.7	IF(Merger_Model!R$150<Close_Date,			
161		Goodwill:	*$ M*		777.3	1,541.5	IFERROR(INDEX(Acquirer!$B$124:$AC$158,			
162		Other Intangible Assets:	*$ M*		119.7	1,304.2	MATCH(Merger_Model!$C159,Acquirer!$C$124:$C$158,0),			
163		Deferred Tax Assets (DTAs):	*$ M*		129.1	225.0	MATCH(Merger_Model!R$150,Acquirer!$B$124:$AC$124,0)),0			
164		Other Long-Term Assets:	*$ M*		42.1	58.9	Merger_Model!Q159-R195-R199-R211-R212-R213))			
165		**Total Long-Term Assets:**	*$ M*		**2,179.5**	**4,920.3**	**4,924.7**	**4,946.0**	**4,976.4**	**5,008.5**

The second case is a simple INDEX/MATCH with an IFERROR wrapped around it for the Net PP&E line from the Acquirer's Balance Sheet on the matching date.

The third case takes the 4Q 20 Net PP&E numbers and subtracts the corresponding CFS lines, such as CapEx and D&A (this subtraction turns CapEx positive and D&A negative).

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Stub Periods and "Rollforwards"

If a deal closes on an **irregular date**, such as August 14th or something else that is not the end of a quarter or fiscal year, and you need to model it accurately, you must create **two "stub period financials"** for the Acquirer and Target.

In this example, we'll walk through a deal that closed on April 30th vs. December 31st fiscal years for both companies.

If you already have 3-statement models for the Acquirer and Target, **Step 1** is to calculate the pre- and post-transaction stub period fractions. In this case, they're ~0.33 and ~0.67 since April 30th is about 1/3 of the fiscal year.

Next, you create "stub" Income Statements and Cash Flow Statements for both periods by multiplying the full-year numbers by these fractions (Acquirer's numbers shown below):

Merger Model - Stub Period Projections for the Acquirer (Fortum Oyj)
(€ in Millions Except Per Share and Per Unit Data)

		Historical:			Projected:	
Acquirer - Income Statement:	***Units:***	**FY17**	**FY18**	**FY19**	**2020-04-30**	**2020-12-31**
Revenue:						
(+) Power Sales:	*€ M*	€ 3,469	€ 4,324	€ 4,714	=Acquirer!$H118*Pre_Txn_Stub	
(+) Heat Sales:	*€ M*	781	797	760	196	398
(+) Other Sales:	*€ M*	740	762	889	272	550
(-) Netting of Nord Pool Transactions:	*€ M*	(367)	(516)	(529)	(143)	(289)
(-) Eliminations:	*€ M*	(103)	(125)	(387)	(60)	(121)
Total Revenue:	*€ M*	**4,520**	**5,242**	**5,447**	**1,549**	**3,137**

The **Balance Sheet** is trickier because the April 30th numbers must be "in-between" the December 31st numbers from the previous year and the current year.

There are two main cases:

1. **Flows in from Cash Flow Statement Stub** – This applies to items like Cash, PP&E, and Debt. Simply link the old version of the item to the corresponding lines on the CFS stub.

2. **Projected with Stub Periods** – Many Working Capital items fall into this category. Take the old number and add (Difference Between Previous and Current FY Numbers) * Pre-Transaction Stub Period Fraction to get the April 30th numbers.

   Here's an example for AR:

		Historical:			Projected:	
Acquirer - Balance Sheet:	***Units:***	**FY17**	**FY18**	**FY19**	**2020-04-30**	**2020-12-31**
ASSETS:						
Current Assets:						
Cash & Cash Equivalents:	*€ M*	€ 3,897	€ 585	€ 1,432	€ 1,551	€ 1,793
Accounts Receivable:	*€ M*	997	1,620	1,176	=G44+(I44-G44)*Pre_Txn_Stub	
Inventory & Other Current Assets:	*€ M*	628	731	494	462	398
Total Current Assets:	*€ M*	**5,522**	**2,936**	**3,102**	**3,134**	**3,199**

This creates a "rolled forward" version of the Balance Sheet as of the transaction close date.

Since the **close date** in this model is variable, the Cash/Debt/Stock mix used to fund the deal varies as the date changes. However, the deal's close date is limited to a 12-month range here rather than the 2+ year range in the previous example, so the funding won't vary nearly as much.

When **combining the statements**, you make all the usual adjustments on the Balance Sheet based on the April 30th version: Cash, Debt, and Stock change, and there are asset write-ups, new Goodwill, and a write-down of the Target's Common Shareholders' Equity.

On the Income Statement and Cash Flow Statement, you will most likely **ignore** the pre-transaction stub through April 30th and just show the combined numbers in the post-transaction stub from May 1st to December 31st, followed by the full combined years after that:

				Comb. Stub:	Projected - Combined Period:			
Combined Income Statement:	Units:		2020-04-30	2020-12-31	FY21	FY22	FY23	FY24
Total Revenue:	€ M			=Acquirer_Stubs!I13+ Target_Stubs!I13			€ 86,831	€ 91,801
Other Income and Operating Expenses:								
(+) Other Income:	€ M			70	112	118	123	126
(-) Materials and Services:	€ M			(37,871)	(66,616)	(73,577)	(79,092)	(83,445
(-) Employee Benefits:	€ M			(903)	(1,505)	(1,609)	(1,703)	(1,791
(+) Expense Synergies:	€ M				-	25	50	50
(-) Depreciation and Amortisation:	€ M			(1,566)	(2,582)	(2,770)	(2,940)	(3,100
(-) Depreciation and Amortisation on Write-Ups:	€ M			(89)	(132)	(132)	(132)	(132
(-) Merger & Integration Costs:	€ M				-	(33)	(17)	-
(-) Other Expenses:	€ M			(315)	(566)	(607)	(650)	(694
(-) Non-Recurring Charges:	€ M			-	-	-	-	-
Total Other Income and Operating Expenses:	€ M			(40,674)	(71,288)	(78,586)	(84,362)	(88,986

Acquirer Stub + Target Stub; full-year acquisition effects are multiplied by 2/3.

Full-year Acquirer + Target financials and full-year acquisition effects.

The **stub periods** do not change any of the conclusions in this model because the EPS accretion/dilution and other analyses are usually based on the **full combined years**.

The IRR would shift due to the timing differences, but it would not make a huge difference unless the "split year" (FY 20) differed greatly from all the other projected years.

These points explain why quarterly models, stub periods, rollforwards, and variable close dates are "on the job" topics: They barely make a difference in most normal scenarios.

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Exchange Ratios and Collars

When a Buyer funds an M&A deal with Stock, there's **inherent ambiguity** about the purchase price. For example, consider the following scenario:

- **Equity Purchase Price:** $1,000
- **Funding:** 100% Stock
- **Buyer's Current Share Price:** $10.00

Based on this, you conclude that the Buyer must issue 100 new shares to do this deal.

But what if the Buyer's share price falls to $9.00?

Does that mean:

a) The Equity Purchase Price **falls to $900** since the Buyer now issues 100 shares worth $9.00 each.

b) The Equity Purchase Price **stays at $1,000**, but the Buyer now issues 111.11 shares rather than 100 shares.

It could be either one – or even a combination of both.

Option A is called a **Fixed Exchange Ratio** and means that **a constant number of shares** is always issued in the deal.

Option B is called a **Floating Exchange Ratio** and means the **Equity Purchase Price stays the same**, so the shares issued change as the Buyer's share price changes.

The Buyer tends to favor the **Fixed Exchange Ratio** if it wants to limit dilution and prefers certainty around the post-deal ownership.

The Seller tends to favor the **Floating Exchange Ratio** if it believes the Buyer's stock price will fall, as this method "locks in" its price and produces a higher ownership percentage if the Buyer's stock price falls.

To compromise, the Buyer and Seller can use a **collar** to establish a Fixed Exchange Ratio within certain share-price ranges and a Floating Exchange Ratio within others.

For example, a **Fixed Exchange Ratio with a collar** might be structured like this:

- **Buyer's Share Price Between $50.00 and $60.00:** The Seller always gets 10 million of the Buyer's shares, so its ownership percentage is fixed, but its effective purchase price ranges from $500 to $600 million.

- **Buyer's Share Price Above $60.00:** The Seller gets a maximum price of $600 million, so it receives fewer shares if the Buyer's share price increases (e.g., 7.5 million shares if the Buyer's share price reaches $80.00).

- **Buyer's Share Price Below $50.00:** The Seller gets a minimum price of $500 million, so it receives more shares if the Buyer's stock price falls (e.g., 12.5 million shares if the Buyer's share price falls to $40.00).

Another option is a **Floating Exchange Ratio with a collar**, which might be structured like this:

- **Buyer's Share Price Between $50.00 and $60.00:** The Seller always gets an effective purchase price of $550 million. So, it receives 11 million shares when the Buyer's share price is $50.00 and ~9.2 million shares when the Buyer's share price is $60.00.

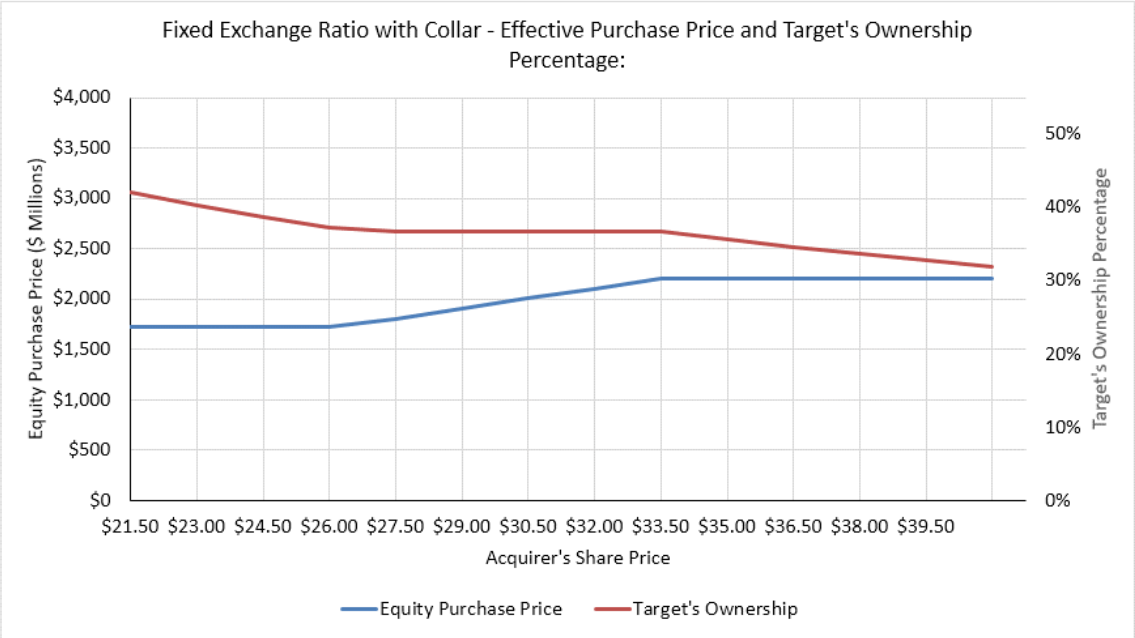
- **Buyer's Share Price Above $60.00:** The Seller always gets ~9.2 million shares, so its ownership stays the same, but its purchase price could climb to much higher levels.

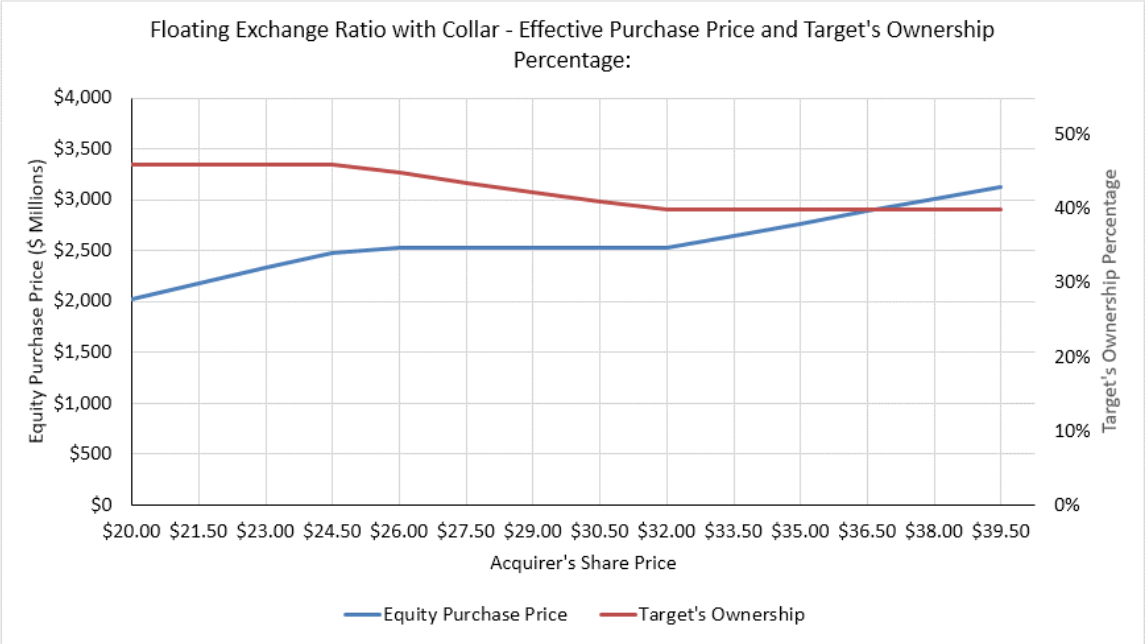
- **Buyer's Share Price Below $50.00:** The Seller always gets 11 million shares, so its ownership stays the same, but its purchase price could fall to much lower levels if the Buyer's share price falls significantly.

If the Seller's main concern is its **ownership** in the combined entity, it could negotiate a Floating Exchange Ratio with a collar to ensure ownership within a specific range.

But if its main concern is the **purchase price**, it might prefer a Fixed Exchange Ratio with a collar to ensure that the purchase price stays within a specific range.

Here's what these structures look like for the Builders FirstSource / BMC deal:





In general, **collars** are most useful when:

- The Seller is **moderately sized** relative to the Buyer – maybe 1 / 4 or 1 / 3 of its size. So, it's not a Merger of Equals, but it's also not a tiny acquisition.

- The parties want the **risk protection** of a Cash deal and the **tax benefits** of a Stock deal.

- It's a **cross-border deal** involving different currencies (collars can also reduce FX risk if the offer price is based on one currency).
- The Buyer's share price has been **volatile,** or the Buyer and Seller strongly disagree about the Buyer's prospects.
- The deal is a **competitive auction**, and one Buyer wants to stand out by offering attractive deal terms.
- The deal will take **a long time to close**. A 100% Stock deal that takes 12 months to close because of regulatory and anti-trust requirements is much riskier than one that takes only 2 – 3 months to close.

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Stock vs. Asset vs. 338(h)(10) Deals

The two main M&A deal structures are **Stock Purchases** and **Asset Purchases**.

In a **Stock Purchase**, the Buyer purchases all the Seller's outstanding shares and gets all its Assets, Liabilities, and off-Balance Sheet items.

In an **Asset Purchase**, the Buyer purchases only selected Assets of the Seller and assumes only selected Liabilities. And it gets only the off-Balance Sheet items listed in the agreement.

100% Cash deals, 100% Debt deals, and 100% Stock deals could all be structured as *either* Stock Purchases *or* Asset Purchases; "Stock" here does not refer to the financing method.

These two structures exist worldwide and work similarly under U.S. GAAP and IFRS, but specific details, such as the tax treatments, may differ from country to country.

Asset Purchases are far more common for divestitures and acquisitions of smaller, private companies; they're difficult to use when acquiring large, public Sellers.

Buyers tend to favor **Asset Purchases** because:

1) They let Buyers pick and choose exactly what they get in deals, which reduces risk; and

2) Assets are written up for *both* Book and Tax purposes, meaning that Buyers can deduct the D&A on Asset Write-Ups for cash-tax purposes.

There are other differences as well: For example, Goodwill amortizes for cash-tax purposes in Asset Purchases, which provides another tax benefit for Buyers.

But in Stock Purchases, Goodwill does not amortize, and the new D&A created in the deal is not cash-tax deductible.

One disadvantage of an Asset Purchase is that the Seller's shareholders must pay taxes based on the purchase price *and* the Gains on its Net Assets.

One disadvantage for the Buyer is that the Seller's Net Operating Losses (NOLs) are written down 100% and may not be used after the deal closes.

Sellers tend to prefer **Stock Purchases** because:

1) Their shareholders pay taxes based on the Purchase Price, which means they are almost always lower because the Gain on Net Assets does not factor in.

2) There's less post-transaction risk because the Buyer acquires everything the Seller has (Assets, Liabilities, and off-Balance Sheet items), so any issues with those items become the Buyer's responsibility.

3) Stock Purchases are faster to execute than Asset Purchases because the Buyer and Seller don't need to specify the treatment of every single Asset and Liability.

To compromise, Buyers and Sellers can use a **338(h)(10) election** to treat a Stock Purchase like an Asset Purchase for tax purposes.

The 338(h)(10) election is specific to the U.S. tax code, so you don't need to know about it if you're in another region. However, similar structures may exist in other countries.

In a 338(h)(10) deal, the Buyer purchases all the Seller's shares and gets all its Assets, Liabilities, and off-Balance Sheet items, but the tax treatment is the same as in an Asset deal.

So, the new D&A on Asset Write-Ups is cash-tax deductible, no new DTL gets created, there's tax-deductible Goodwill amortization, and the Seller's NOLs are written down 100%.

338(h)(10) deals have additional requirements as well; for example, the Buyer must be a C corporation, the Seller must be U.S.-based, and the Buyer must acquire at least 80% of the Seller's "value and voting power" within 12 months.

Here's a summary:

Structure:	Stock Purchase	Asset Purchase	338(h)(10) Election
Buyer Acquires:	All Assets, Liabilities, and Off-BS Items	Only *Certain* Assets and Liabilities	All Assets, Liabilities, and Off-BS Items
Seller Pays Taxes Based On:	Purchase Price	Purchase Price PLUS (Value Assigned to Net Assets in Deal – Book Value of Net Assets)	
Assets Written Up on Tax Balance Sheet?	No	Yes	
Can Buyer Deduct New D&A on Asset Write-Ups for Cash-Tax Purposes?	No	Yes	
Creates New DTL?	Yes	No	
Goodwill & Other Intangibles:	Not amortized for tax purposes and not cash-tax-deductible	Amortization is tax-deductible; amortized over 15 years for cash-tax purposes	
Allowable Annual NOL Usage:	Equity Purchase Price * Maximum of Past 3 Months' Adjusted Long-Term Rates	None	
Off-Balance Sheet NOL Write-Down:	MAX(0, NOL Balance – Allowable Annual Usage * # Years Until Expiration)	100%	
Write-Down of NOL Portion of DTA:	MAX(0, NOL Portion – Allowable Annual Usage * Buyer's Tax Rate * # Years Until Expiration)	100%	
Most Common Sellers:	Public companies and large private companies	Private companies, divested divisions, and distressed public companies	
Favored By:	Sellers	Buyers	Both

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Net Operating Losses (NOLs) in Different Deal Structures

We summarized the treatment of the Seller's NOLs in the table on the previous page, but it's worth explaining it in more detail with a few sample numbers.

In Asset Purchases and 338(h)(10) deals, the Seller's NOLs are written down 100% and cannot be used post-transaction. In Stock Purchases, the Buyer can use *some* of the Seller's NOLs each year, but it may have to write down a portion of the total balance.

Let's say the Seller has $100 million in off-Balance Sheet NOLs, which are represented as $25 million within its Deferred Tax Asset since it has a 25% tax rate.

The Buyer pays an Equity Purchase Price of $1 billion for the Seller.

In an Asset Purchase or 338(h)(10) deal, the Buyer writes down the $25 million of NOLs within the DTA and the entire $100 million off-Balance Sheet number, and it cannot use *any* of the NOLs in the future.

But this full write-down does **not** happen in a Stock Purchase, and the Buyer can use a limited amount of NOLs annually.

For U.S.-based Buyers, the allowable usage is calculated as follows:

- **Allowable Annual NOL Usage** = Equity Purchase Price * Highest of Past 3 Months' Adjusted Long-Term Rates

The rules vary in different countries; in some cases, there is no limit or a much higher annual limit, but the NOLs themselves may also expire more quickly (or not at all).

These "Adjusted Long-Term Rates" are linked to the prevailing yields on government bonds in the Buyer's country (i.e., the Risk-Free Rate).

If the past three months had "Adjusted Long-Term Rates" of 1.2%, 1.0%, and 1.5%, the Buyer could use a maximum of $1 billion * 1.5% = $15 million in NOLs each year.

Therefore, the Buyer could use the Seller's entire $100 million NOL balance over ~7 years to reduce its Cash Taxes each year.

Therefore, if these NOLs do not expire or expire *after* 7 years, no write-down is required because the Buyer can use the full balance before that happens.

As the Buyer uses these NOLs, the off-Balance Sheet figure will decline by $15 million per year, and the portion within the DTA will decline by $15 million * 25% = $3.75 million per year.

However, if these NOLs were set to expire in **3 years**, the Buyer could **not** use the entire balance.

In that case, the Buyer could use only $45 million of the NOLs, corresponding to $11.25 million of the DTA balance.

So, the Buyer, **in the initial transaction**, would have to write down $55 million of the off-Balance Sheet NOL balance and $13.75 million of the DTA balance.

After that initial write-down, the off-Balance Sheet NOLs would decrease by $15 million per year, and the DTA would decrease by $3.75 million per year.

Here's an example of how the Taxable Income and Cash Taxes differ in a Stock vs. Asset or 338(h)(10) deal:

Stock Purchase – Book vs. Cash Taxes

		Projected:			
Book vs. Cash Tax Projections:	*Units:*	FY20	FY21	FY22	FY23
Book Taxable Income:	*$ M*	$ -	$ 393.7	$ 639.3	$ 788.5
Book Tax Expense:	*$ M*	-	(86.6)	(140.6)	(173.5)
Cash Tax Calculation:					
(+) Book Taxable Income:	*$ M*	-	393.7	639.3	788.5
(+) Book Amortization of Goodwill:	*$ M*	-	-	-	-
(+) Book Amortization of New Intangibles:	*$ M*	-	68.5	68.5	68.5
(+) Book Depreciation of PP&E Write-Up:	*$ M*	-	5.2	5.2	5.2
(-) Tax Amortization of Goodwill:	*$ M*	-	-	-	-
(-) Tax Amortization of New Intangibles:	*$ M*	-	-	-	-
(-) Tax Depreciation of PP&E Write-Up:	*$ M*	-	-	-	-
Pre-NOL Cash Taxable Income:	*$ M*	**-**	**467.4**	**712.9**	**862.2**
Beginning NOL Balance:	*$ M*	93.5	77.4	55.9	34.4
(+) NOLs Created:	*$ M*	-	-	-	-
(-) NOLs Used:	*$ M*	-	(21.5)	(21.5)	(21.5)
Ending NOL Balance:	*$ M*	**93.5**	**55.9**	**34.4**	**12.9**
NOL-Adjusted Taxable Income:	*$ M*		445.9	691.4	840.7
Cash Taxes Payable:	*$ M*		(98.1)	(152.1)	(185.0)
Deferred Income Taxes:	*$ M*		**(11.5)**	**(11.5)**	**(11.5)**
Change in DTLs:	*$ M*		(16.2)	(16.2)	(16.2)
Change in DTAs:	*$ M*		4.7	4.7	4.7

Asset / 338(h)(10) Purchase – Book vs. Cash Taxes

		Projected:			
Book vs. Cash Tax Projections:	*Units:*	FY20	FY21	FY22	FY23
Book Taxable Income:	*$ M*	$ -	$ 393.7	$ 639.3	$ 788.5
Book Tax Expense:	*$ M*	-	(86.6)	(140.6)	(173.5)
Cash Tax Calculation:					
(+) Book Taxable Income:	*$ M*	-	393.7	639.3	788.5
(+) Book Amortization of Goodwill:	*$ M*	-	-	-	-
(+) Book Amortization of New Intangibles:	*$ M*	-	68.5	68.5	68.5
(+) Book Depreciation of PP&E Write-Up:	*$ M*	-	5.2	5.2	5.2
(-) Tax Amortization of Goodwill:	*$ M*	-	(36.6)	(36.6)	(36.6)
(-) Tax Amortization of New Intangibles:	*$ M*	-	(22.8)	(22.8)	(22.8)
(-) Tax Depreciation of PP&E Write-Up:	*$ M*	-	(7.0)	(7.0)	(7.0)
Pre-NOL Cash Taxable Income:	*$ M*	**-**	**401.0**	**646.5**	**795.8**
Beginning NOL Balance:	*$ M*	-	-	-	-
(+) NOLs Created:	*$ M*	-	-	-	-
(-) NOLs Used:	*$ M*	-	-	-	-
Ending NOL Balance:	*$ M*	**-**	**-**	**-**	**-**
NOL-Adjusted Taxable Income:	*$ M*		401.0	646.5	795.8
Cash Taxes Payable:	*$ M*		(88.2)	(142.2)	(175.1)
Deferred Income Taxes:	*$ M*		**(1.6)**	**(1.6)**	**(1.6)**
Change in DTLs:	*$ M*		(1.6)	(1.6)	(1.6)
Change in DTAs:	*$ M*		-	-	-

In this example, the ending Cash balance is higher in the Asset or 338(h)(10) deal because the Buyer can deduct the new D&A for cash-tax purposes.

The Seller's NOLs are written down, which hurts the Buyer, but the tax advantages of the D&A treatment outweigh this loss.

If the Seller has a huge NOL balance, a Stock Purchase might be superior, but if the NOL balance is small or non-existent, an Asset Purchase or 338(h)(10) election might offer more benefits for the Buyer.

The exact structure rarely makes a big difference in a merger model unless the deal parameters are unusual (e.g., the Seller has a massive amount of NOLs, or the Seller's Assets are written up by huge percentages).

But it's still good to be familiar with these deal structures because they come up in more advanced interviews and on the job.

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Interview Questions

It is unlikely that you will get questions about these more advanced M&A concepts in interviews.

However, if you've had significant work experience or presented a full list of M&A deals on your resume/CV, these more advanced questions have a higher probability of coming up.

Quarterly Models, Variable Close Dates, and Stub Periods

Questions about "timing differences" are much more likely to come up in the context of **deal discussions** around the models and technical work you completed, but interviewers could still quiz you on the basics in standalone questions.

1. How does a quarterly merger model differ from an annual one? Does it add any value?

Most of the differences concern the financial projections for the Buyer and Seller; in a quarterly model, you use Year-Over-Year (YoY) growth rates for revenue and expenses, you use LTM Income Statement figures to forecast the Balance Sheet, and you use quarterly versions of the interest rates and repayment terms for Debt. Annual summaries and roll-ups are also common.

In the merger model itself, you often allow the deal to close on different quarterly end dates so you can compare the accretion/dilution in different scenarios.

Quarterly merger models are most useful when there's significant uncertainty around the deal's close date, and at least one company changes significantly over time. Outside of that, the added precision does not necessarily justify the extra time and effort required.

2. Suppose you have built a quarterly merger model with a variable close date. The "baseline" close date is currently Quarter 2 of Year 1. If the deal closes later, such as in Quarter 2 of Year 2, how would the EPS accretion/dilution change?

It depends on how the Buyer and Seller's financial profiles change over this period. If their Cash balances are higher and their Debt balances are lower, the deal is likely to be more accretive because it means they can use more Cash and Debt to fund the deal, which is usually cheaper than Stock.

On the other hand, if these balances move in the wrong direction due to higher burn rates or reduced cash flows, a later close date would be more dilutive because it means the Stock component will be greater.

3. Is that answer complete? Can you answer this question based on Excel and the financial statements? What else factors in?

The previous answer is all you can say *based on the financial statements and the Excel model*.

However, in real life, the other major factor is that the Buyer and Seller's share prices often change significantly after the deal is announced, and these movements also affect the accretion/dilution if the deal has a Stock component.

If the Buyer's share price increases, any deal with a Stock component becomes more accretive, while the opposite happens if its share price decreases (though this varies based on how the Exchange Ratio is set, possible collars, etc.).

4. What are the main "cases" when combining the Buyer and Seller's financial statements in a merger model with variable close dates?

The main cases on the Income Statement and Cash Flow Statement are "pre-transaction" and "post-transaction." The pre-transaction numbers are only the Buyer's, while the post-transaction ones include the Buyer + Seller's numbers *and* all the acquisition effects, adjusted based on the periods (monthly vs. quarterly vs. annual) and the possible close date.

On the Balance Sheet, there's also the "close date" case, which is different since the Balance Sheet suddenly changes to reflect the Cash, Debt, and Stock used in the deal, the new Goodwill, the new DTL, the asset write-ups, and so on.

The Balance Sheet is trickier than the other statements because many of the line items require different formulas in the post-transaction period; for example, the Working Capital lines are simple combinations of the Buyer and Seller, but items such as Net PP&E and Debt flow in from the Combined Cash Flow Statement.

5. What would change in a merger model if the deal closed on an irregular date, such as August 15th?

You would "roll forward" the Balance Sheets for both companies to August 15th and combine them on that date, ensuring that the Purchase Price Allocation and Sources & Uses schedules are also based on that date.

You would also create "stub periods" for the Combined Income Statement and Cash Flow Statement to show what happens between the most recent period before August 15th and August 15th and what happens between August 15th and the *next* quarter (or year) after that.

In the merger model, you would combine the statements starting in this second stub period (August 15th to the end of the next quarter/year), but you would still focus on the **first full year of combined results** because EPS accretion/dilution is more meaningful over a full year.

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Exchange Ratios and Collars

Questions about these topics are often conceptual, but it's possible to receive a few "mental math" questions about the impact of Exchange Ratios and collars on the shares issued in a deal and the EPS accretion/dilution.

1. What's the difference between Fixed and Floating Exchange Ratios, and which do Buyers prefer?

With a Fixed Exchange Ratio, the Seller receives a constant number of shares regardless of the Buyer's share price. For example, the Buyer might agree to issue two new shares for each of the Seller's shares. The Seller's ownership in the combined company stays the same as the Buyer's share price changes, but its purchase price will change.

With a Floating Exchange Ratio, the Seller receives a fixed purchase price but a variable number of shares. For example, the Buyer might agree to pay $200 million to the Seller, but that means 10 million shares if the Buyer's share price is $20.00 and 40 million shares if the Buyer's share price is $5.00.

Buyers care the most about **avoiding dilution** in M&A deals, so they tend to favor the Fixed Exchange Ratio if they're not confident of their current/future share price. If the Buyer is reasonably confident that its share price will rise, it might favor a Floating Exchange Ratio so that it issues fewer shares.

2. Why might a Buyer and Seller agree to a collar in a 100% Stock deal?

100% Stock deals present risk for both Buyers and Sellers: The Buyer could dilute its shareholders by a huge amount if its share price falls, while the Seller could receive fewer shares than expected if the Buyer's share price rises.

A **collar** allows for a compromise and reduces the risks by establishing a Fixed Exchange Ratio within a certain range of Buyer share price, with Floating Exchange Ratios outside that region. It might also be set up in the opposite way, with a Floating Exchange Ratio in a certain range and Fixed Exchange Ratios outside that range.

Sellers can be assured of receiving a fixed purchase price or a fixed number of shares within a range of share prices for the Buyer, while Buyers can limit either the dilution or the effective purchase price.

3. What are example terms for a Fixed Exchange Ratio with a collar in an M&A deal?

This structure means that the Seller gets a fixed number of shares within a certain share price range for the Buyer.

So, the purchase price will vary within that range, and above or below that range, the purchase price is fixed, but the shares issued will vary. For example:

- **Buyer's Share Price Between $50.00 and $60.00:** The Seller always gets 10 million of the Buyer's shares ($500 to $600 million purchase price).
- **Buyer's Share Price Above $60.00:** The Seller always gets a $600 million purchase price, but the shares issued vary based on the Buyer's share price.
- **Buyer's Share Price Below $50.00:** The Seller always gets a $500 million purchase price, but the shares issued vary based on the Buyer's share price.

4. How does EPS accretion/dilution change with a Fixed vs. Floating Exchange Ratio? What if there's a collar?

A Fixed Exchange Ratio does not impact the accretion/dilution because the shares issued by the Buyer are constant regardless of its share price. But with a Floating Exchange Ratio, the shares issued change as the Buyer's share price changes, so the deal will become more accretive as the Buyer's share price increases and more dilutive as it falls.

With a collar, the EPS accretion/dilution changes *over only specific ranges* of the Buyer's share price. So, it might vary between $50.00 and $70.00 per share but remain constant below $50.00 and above $70.00 if the Exchange Ratio is Fixed in those ranges (for example).

5. Suppose that a Buyer plans to pay $500 million for a Seller using 100% Stock. Its current share price is $50.00, so it will issue 10 million shares to do the deal.

If the deal is based on a Floating Exchange Ratio between $40.00 and $60.00 with a collar outside those levels, what are the minimum and maximum purchase prices and shares issued?

There are **no** minimum or maximum purchase prices when there's a Floating Exchange Ratio with a collar; the *shares issued* are capped above $60.00 and below $40.00, but the purchase price keeps changing.

The purchase price here will be $500 million if the Buyer's share price stays between $40.00 and $60.00; if it goes outside that range, the purchase price will change.

The shares issued will be between $500 million / $60.00 = 8.33 million and $500 million / $40.00 = 12.5 million. These numbers represent the minimum and maximum number of shares issued due to this collar structure.

6. When is a collar most and least useful?

A collar is *most useful* in a 100% Stock deal when the Seller is moderately sized relative to the Buyer (e.g., 1 / 4 or 1 / 3 of its size), the parties want to compromise and reduce their risk, and there's disagreement about the Buyer's share price or prospects. Any deal expected to take 12+ months to close may also benefit from a collar due to the risk of share price volatility before the close.

A collar is *least useful* when most of these conditions are not true (e.g., there's little risk of share price volatility or the Buyer is 100x bigger than the Seller).

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Stock vs. Asset vs. 338(h)(10) Deals and NOLs

These questions start to border on “obscure,” but they may come up in more advanced interviews with M&A groups, especially in the U.S.

1. Why do Buyers tend to prefer Asset Purchases, while Sellers tend to prefer Stock Purchases?

In Asset Purchases, Buyers can pick and choose the exact Assets they want to acquire and the exact Liabilities they want to assume, which reduces the transaction risk.

Buyers can also deduct D&A on Asset Write-Ups for cash-tax purposes in an Asset Purchase, which reduces their tax burden after the deal closes.

Sellers tend to prefer Stock Purchases because Asset Purchases leave them with more risk after deals close and because they must pay taxes based on the purchase price PLUS the Gains recorded on their Net Assets in an Asset Purchase.

2. What's the advantage of a 338(h)(10) election for a U.S.-based Buyer?

In a 338(h)(10) Election, the Buyer and Seller treat a Stock Purchase as an Asset Purchase for tax purposes.

So, the Buyer still acquires all the Assets, Liabilities, and off-Balance Sheet items of the Seller, but it **can** also deduct D&A on Asset Write-Ups for cash-tax purposes.

The Seller’s entire NOL balance is also written down (as in a standard Asset Purchase).

338(h)(10) deals can help Buyers and Sellers compromise and reach an agreement more quickly since they combine the elements favored by Buyers in Asset Purchases and Sellers in Stock Purchases.

3. If a Seller has a massive NOL balance, should the Buyer use a Stock Purchase, Asset Purchase, or 338(h)(10) election to acquire it?

The Buyer should use a Stock Purchase because NOLs are written down 100% in Asset Purchase and 338(h)(10) deals, so the Buyer cannot use the Seller’s NOLs in those deal structures.

4. Walk me through what happens in a Stock Purchase deal where the Buyer pays an Equity Purchase Price of $2 billion for the Seller, and the Seller has an off-Balance Sheet NOL balance of $400 million. The NOLs expire in 5 years.

Assume that the Long-Term Adjusted Rates for the past three months were 0.5%, 0.7%, and 1.0% and that the Buyer's Tax Rate is 25%.

If the off-BS NOL balance is $400 million, the portion within the DTA should be approximately $100 million at a 25% tax rate.

The Buyer can use MAX(0.5%, 0.7%, 1.0%) * $2 billion, or $20 million, of NOLs per year.

The NOLs expire in 5 years, which means the Buyer can use 5 * $20 million = $100 million total.

Therefore, the Buyer will write down $300 million of the off-BS NOLs and $75 million of the NOLs within the DTA when the transaction closes.

The remaining off-BS NOL balance will be $100 million, and the NOL portion within the DTA will be $25 million.

After the deal closes, the Buyer will use $20 million of the NOLs each year to reduce its cash-taxable income, so the off-BS balance will decline by $20 million per year.

The DTA portion will decline by $5 million per year until both the on-BS and off-BS NOL balances reach $0 at the end of Year 5.

5. How do these numbers change in an Asset Purchase?

In an Asset Purchase, the Seller's Net Operating Losses – both the off-Balance Sheet and on-Balance Sheet versions – are written down to $0, and the Buyer can't use any of the Seller's NOLs.

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